

# IEEE 830

**Software Requirements Specification**

**For**

**Home Network Security Monitoring Dashboard**

**Version 3**

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## Revision History

| Name            | Date       | Reason for Changes                                            | Version |
|-----------------|------------|---------------------------------------------------------------|---------|
| Jakub Kasztelan | 10.03.2026 | Initial version                                               | V 1     |
| Jakub Kasztelan | 23.03.2026 | Created Use Case Diagram, altered and specified functionality | V 2     |
| Jakub Kasztelan | 14.04.2026 | Added Class Diagram                                           | V 3     |

## **1. Introduction**

### **1.1. Purpose**

The purpose of the program is to organize the home network security scanning into a dashboard.

### **1.2. Intended Audience and Reading Suggestions**

The system is intended for people with every level of knowledge on networking and security.

## **2. Overall Description**

### **2.1. Product Perspective**

The system is a new, stand-alone product. It shall work independently of other products.

### **2.2. Product Features**

- Scanning the home network for inbound traffic.
- Providing scan analysis.
- Highlighting suspicious traffic.
- Generating a scan report.

### **2.3. User Classes and Characteristics**

«Identify the various user classes (intended target for the product) that you anticipate will use this product. User classes may be differentiated based on frequency of use, subset of product functions used, technical expertise, security or privilege levels, educational level, or experience. Describe the pertinent characteristics of each user class. Certain requirements may pertain only to certain user classes. Distinguish the most important user classes for this product from those who are less important to satisfy.»

### **2.4. Operating Environment**

«Describe the environment in which the software will operate, including the hardware platform, operating system and versions, and any other software components or applications with which it must peacefully coexist.»

### **2.5. Design and Implementation Constraints**

«Describe any items or issues that will limit the options available to the developers. These might include: corporate or regulatory policies; hardware limitations (timing requirements, memory requirements); interfaces to other applications; specific technologies, tools, and databases to be used; parallel operations; language requirements; communications protocols; security considerations; design conventions or programming standards (for example, if the customer's organization will be responsible for maintaining the delivered software).»

### **2.6. User Documentation**

«List the user documentation components (such as user manuals, on-line help, and tutorials) that will be delivered along with the software. Identify any known user documentation delivery formats or standards.»

### **2.7. Assumptions and Dependencies**

«List any assumed factors (as opposed to known facts) that could affect the requirements stated in the specification. These could include third-party or commercial components that you plan to use, issues around the development or operating environment, or constraints. The project could be affected if these assumptions are incorrect, are not shared, or change. Also identify any dependencies the project has on external factors, such as software components that you intend to reuse from another project, unless they are already documented elsewhere (for example, in the vision and scope document or the project plan).»

## **3. System Features**

«The purpose here is to give a brief introduction to the system features, so that §4 has something to reference. Then, the full explanation of the system features is given in §5, likely referencing §4. The numbering in §3 and §5 should match.»

### **3.1. «Feature Name»**

«Brief description of the feature, generally 1-3 sentences»

### **3.«N». «Additional feature blocks as needed»**

## **4. External Interface Requirements**

### **4.1. User Interfaces Overview**

The system shall provide a web-based GUI, accessible from a standard browser. It shall be composed of:

- Main Dashboard: shall display real-time network security status.
- Traffic Visualization: shall display the chart of inbound traffic and most active external hosts. Shall display the inbound network traffic in a user-friendly way and highlight any suspicious activity.
- Control Panel: shall display buttons to start/stop scanning and export a report.

### **4.2. Hardware Interfaces**

«Describe the logical and physical characteristics of each interface between the software product and the hardware components of the system. This may include the supported device types, the nature of the data and control interactions between the software and the hardware, and communication protocols to be used.»

### **4.3. Software Interfaces**

«Describe the connections between this product and other specific software components (name and version), including databases, interpreters, operating systems, tools, libraries, and integrated commercial components. Identify the data items or messages coming into the system and going out and describe the purpose of each. Describe the services needed and the nature of communications. Identify data that will be shared across software components. If the data sharing mechanism must be implemented in a specific way (for example, use of a global data area in a multitasking operating system), specify this as an implementation constraint.»

## **5. System Features/Modules**

«This template illustrates organizing the functional requirements for the product by system features, the major services provided by the product.»

### **5.1. Home Network Scanning**

#### **5.1.1 Description and Priority**

«Provide a short description of the feature. Give its priority, too.»

#### **5.1.2 Stimulus/Response Sequences**

«"Stimulus" or "Condition": «Describe the stimulus or condition»

Response: «Describe the response»

«Additional Stimulus/Response pairs as needed»

#### **5.1.3 Functional Requirements**

«Itemize the detailed functional requirements associated with this feature. These are the software capabilities that must be present in order for the user to carry out the services provided by the feature. Include how the product should respond to anticipated error conditions or invalid inputs. Requirements should be concise, complete,

unambiguous, verifiable, and necessary. Each requirement should be uniquely identified with a sequence number or a meaningful tag of some kind.»

FR-01: The system shall scan the home network in real time for inbound traffic.

FR-02: The system shall allow the user to start/stop scanning of the network.

## **5.2. Home Network Scan Analysis**

### **5.2.3 Functional Requirements**

FR-01: The system shall provide a network traffic summary (Inbound traffic hosts, types of performed operations)

FR-02: The system shall highlight suspicious network activity.

FR-03: The system shall allow the user to download traffic report.

## **6. Nonfunctional Requirements**

### **6.1. Performance**

NFR-01: The system shall maintain a <time (seconds)> scan delay.

NFR-02: The system shall use <encryption type> encryption.

NFR-03: The system shall maintain <proc> uptime.

NF-1.«N»: «Any performance-related issue»

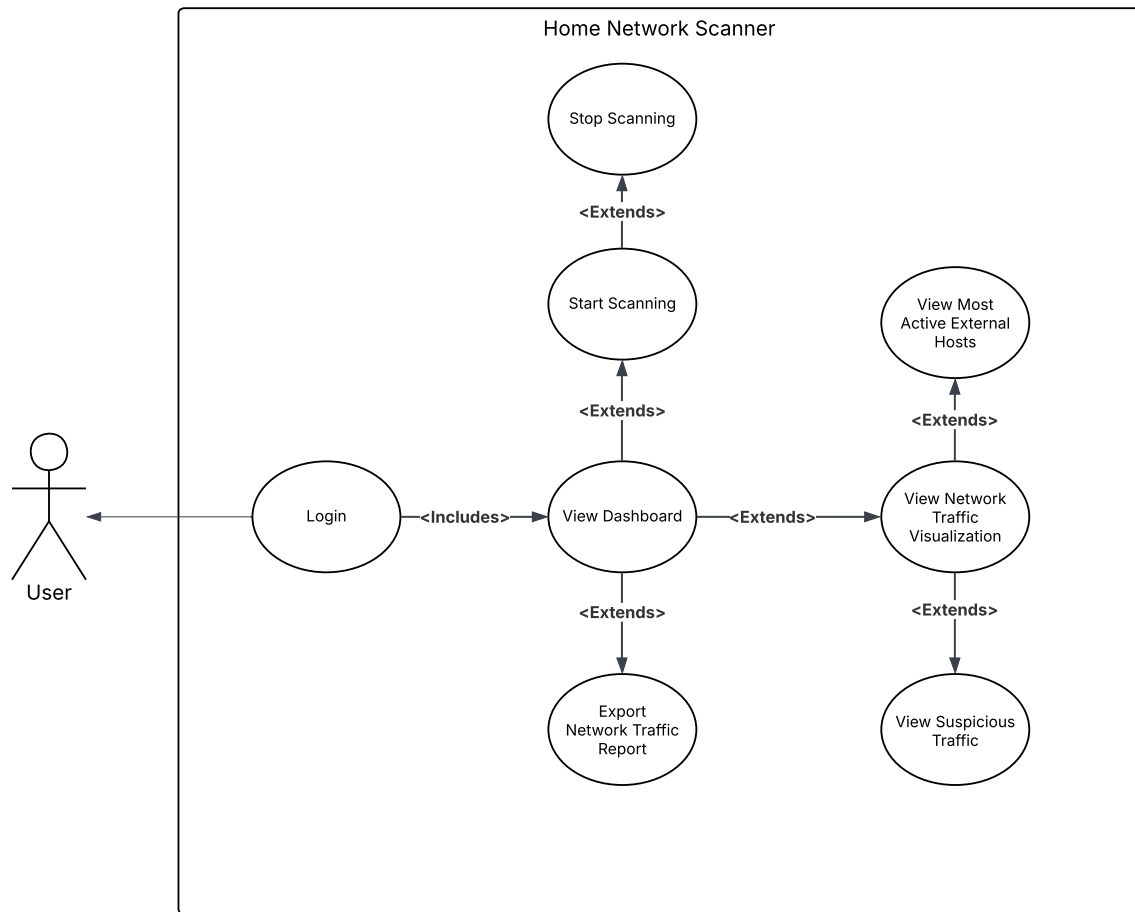
### **6.2. Security**

NF-2.1: «Any security-related issue»

NF-2.«N»: «Any security-related issue»

6.«N». «Any additional “nonfunctional” blocks»

## 7. Use Case Diagram



## 8. Class Diagram

